

Challenges-4

Use Case: Setting up Transit Gateway and VPC Endpoints for a Multi-VPC Architecture

Scenario:

A large organization is migrating its on-premises infrastructure to the AWS cloud. The organization's architecture involves multiple VPCs for different departments and applications, each requiring secure communication with centralized services and external resources.

The IT team needs to design and implement a scalable and efficient network architecture to accommodate the organization's growth and ensure robust connectivity between VPCs and external services.

Objectives:

- Design and deploy a scalable network architecture using AWS Transit Gateway to simplify network connectivity between multiple VPCs.
- Configure VPC endpoints to securely access AWS services without internet gateways or NAT gateways, ensuring data privacy and minimizing exposure to external threats.

Objectives

- Use Transit Gateway to simplify multi-VPC connectivity.
- Configure VPC Endpoints for secure service access without internet exposure.

Steps to Implement

1. Create Multiple VPCs with unique CIDRs (e.g., `10.10.0.0/16`, `10.20.0.0/16`).
2. Create Transit Gateway (`Org-TGW`).
3. Attach VPCs to TGW.
4. Update Route Tables in each VPC to route traffic via TGW.
5. Configure VPC Endpoints (Gateway for S3/DynamoDB, Interface for EC2 API/CloudWatch).
6. Launch EC2 Instances in each VPC (public + private subnets).
7. Test Connectivity:
 - Ping EC2s across VPCs (TGW).
 - Run `aws s3 ls` from private EC2 (via Endpoint).

Result

- Transit Gateway provides centralized routing between VPCs.
- VPC Endpoints allow secure, private access to AWS services.
- Architecture is scalable, secure, and enterprise-ready.



Step-by-Step AWS Networking Lab

1. Create Multiple VPCs

- Go to VPC → Create VPC.
- Make at least 2–3 VPCs (e.g., **DeptA-VPC**, **DeptB-VPC**, **Shared-Services-VPC**).
- Each VPC:
 - CIDR block (e.g., **10.10.0.0/16**, **10.20.0.0/16**, **10.30.0.0/16**).
 - Create public + private subnets in each.

2. Create Transit Gateway

- Go to VPC → Transit Gateways → Create Transit Gateway.
- Name: **Org-TGW**.
- Leave ASN default (64512).
- Enable default route table association/propagation.
- Status: Available .

3. Attach VPCs to TGW

- Go to Transit Gateway Attachments → Create Attachment.
- Select TGW: **Org-TGW**.
- Choose VPC: **DeptA-VPC**.
- Select subnets (one per AZ).
- Repeat for all VPCs.
- Attachments show Pending → Available.

4. Update Route Tables

- In each VPC's route table:
 - Add route → Destination: CIDR of other VPC.
 - Target: Transit Gateway (**tgw-xxxx**).
- Example:
 - In **DeptA-VPC**: route **10.20.0.0/16** → **TGW**.
 - In **DeptB-VPC**: route **10.10.0.0/16** → **TGW**.

5. Configure VPC Endpoints

- Go to VPC → Endpoints → Create Endpoint.
- Choose service:
 - Gateway Endpoint → S3, DynamoDB.
 - Interface Endpoint → EC2 API, CloudWatch, etc.
- Select VPC + private subnets.
- Attach security group allowing traffic from EC2s.
- AWS auto-updates route tables for Gateway Endpoints.

6. Launch EC2 Instances

- Launch one EC2 per VPC.
- Public subnet → enable auto-assign public IP (for SSH).
- Private subnet → no public IP (for TGW + Endpoint testing).
- Security Groups:
 - Allow inbound SSH (22) from your laptop IP.
 - Allow ICMP (ping) from peer VPC CIDRs.

7. Test Connectivity

1. SSH into EC2 in public subnet.
2. Ping EC2 in another VPC (via TGW).

From private EC2 → run:

```
bash
```

```
aws s3 ls
```

3. 👉 Should work via VPC Endpoint without IGW/NAT.

✓ Final Architecture

- **Transit Gateway = central router connecting all VPCs.**
- **VPC Endpoints = secure access to AWS services without internet.**
- **EC2s = test connectivity across VPCs and to AWS services.**

The Network

Network	CIDR	Purpose
Production	VPC 10.10.0.0/16	Production applications
Development	VPC 10.20.0.0/16	Development applications
Shared Services	VPC 10.30.0.0/16	DNS, monitoring, security, etc.
Default	172.31.0.0/16	Existing network

Created the VPCs

VPC - 1

Name: prod-vpc

IPv4 CIDR: 10.10.0.0/16

VPC - 2

Name: dev-vpc

IPv4 CIDR: 10.20.0.0/16

VPC - 3

Name: shared-services-vpc

IPv4 CIDR: 10.30.0.0/16

Created Subnets

prod-vpc

private-subnet-a 10.10.11.0/24

dev-vpc

private-subnet-a 10.20.11.0/24

shared-services-vpc

private-subnet-a 10.30.11.0/24

Created the Transit Gateway:

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

Your VPCs

VPCsVPC encryption controls

Your VPCs (1/4) Info

Find VPCs by attribute or tag

	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR
☐	-	vpc-09ace864d17ee9449	Available	-	-	Off	172.31.0.0/16
☐	DeptB-VPC	vpc-0340797b746c93cf3	Available	-	-	Off	10.20.0.0/16
☒	DeptA-VPC	vpc-06832a564ba035c30	Available	-	-	Off	10.10.0.0/16
☐	Shared-Services-VPC	vpc-0980cb3ded048bde4	Available	-	-	Off	10.30.0.0/16

vpc-06832a564ba035c30 / DeptA-VPC

Details

Resource map

CIDRs

Flow logs

Tags

Related resources

Integrations

Details

VPC ID

State

Block Public Access

DNS hostnames

DNS resolution

Tenancy

DHCP option set

Main route table

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

Subnets

Subnets (1/12) Info

Find subnets by attribute or tag

	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 C
☐	DeptB-Public-Subnet	subnet-00f68524e99c2fd63	Available	vpc-0340797b746c93cf3 Dept...	Off	10.20.1.0/24	-
☐	DeptB-Private-Subnet	subnet-0da7b4e6d43387637	Available	vpc-0340797b746c93cf3 Dept...	Off	10.20.2.0/24	-
☐	-	subnet-0e2d3f014dfee4a12	Available	vpc-09ace864d17ee9449	Off	172.31.16.0/20	-
☐	-	subnet-0044d4f9336cfeedf	Available	vpc-09ace864d17ee9449	Off	172.31.0.0/20	-
☒	DeptA-Public-Subnet	subnet-020f97b1eb9cf5783	Available	vpc-06832a564ba035c30 Dep...	Off	10.10.1.0/24	-
☐	-	subnet-092952d3a2e3cb464	Available	vpc-09ace864d17ee9449	Off	172.31.32.0/20	-
☐	-	subnet-022d57519c75dd10e	Available	vpc-09ace864d17ee9449	Off	172.31.64.0/20	-
☐	-	subnet-0403739edceea0532	Available	vpc-09ace864d17ee9449	Off	172.31.48.0/20	-

subnet-020f97b1eb9cf5783 / DeptA-Public-Subnet

Details

Flow logs

Route table

Network ACL

CIDR reservations

Sharing

Tags

Related resources

Details

Subnet ID

Subnet ARN

State

Block Public Access

IPv4 CIDR

IPv6 CIDR association ID

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

Route tables

Route tables (1/10) Info

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Own.
☐	-	rtb-0b744ec31e7b34d60	-	-	Yes	vpc-0340797b746c93cf3 DeptB-VPC	9868
☐	DeptB-pub-RT	rtb-05bbca45f934ef75bb	subnet-00f68524e99c2fd...	-	No	vpc-0340797b746c93cf3 DeptB-VPC	9868
☐	DeptB-pri-RT	rtb-0ecaf4f41b6060856	subnet-0da7b4e6d43387...	-	No	vpc-0340797b746c93cf3 DeptB-VPC	9868
☐	-	rtb-0b846f22a3769159	-	Yes	Yes	vpc-06832a564ba035c30 DeptA-VPC	9868
☐	DeptA-pub-RT	rtb-08b162ef91194d422	subnet-020f97b1eb9cf57...	-	No	vpc-06832a564ba035c30 DeptA-VPC	9868
☐	DeptA-pri-RT	rtb-0cbbfd088215dd5c7	subnet-05baff6d1c85dda...	-	No	vpc-06832a564ba035c30 DeptA-VPC	9868
☐	-	rtb-0adcd239fd379eb06	-	Yes	Yes	vpc-0980cb3ded048bde4 Shared-Services...	9868
☒	Shared-Pri-RT	rtb-06a6892a2ed840d07	subnet-076054c1090ae5...	-	No	vpc-0980cb3ded048bde4 Shared-Services...	9868

rtb-06a6892a2ed840d07 / Shared-Pri-RT

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Details

Route table ID

Main

Explicit subnet associations

Edge associations

VPC

Owner ID

aws

Search

[Alt+S] Ask Amazon Q

United States (N. Virginia)

chandrashekar (986881030460)

chandrashekar

VPC > Internet gateways

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

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Security

Network ACLs

Security groups

Internet gateways (4)

Find internet gateways by attribute or tag

☐	Name	Internet gateway ID	State	VPC ID	Owner
☐	DeptA-IGW	igw-06fe626f221977f1	Attached	vpc-06832a564ba035c30 DeptA-VPC	986881030460
☐	Shared-igw	igw-085a7feacbac3830	Attached	vpc-0980cb3ded048bde4 Shared-Serv...	986881030460
☐	-	igw-09c3346549babe6f4	Attached	vpc-09acc864d17ee9449	986881030460
☐	DeptB-IGW	igw-0ab0be6fcec2d7174	Attached	vpc-0340797b746c93cf3 DeptB-VPC	986881030460

Select an internet gateway above

aws

Search

[Alt+S] Ask Amazon Q

United States (N. Virginia)

chandrashekar (986881030460)

chandrashekar

VPC > Transit gateways

providers

Verified Access groups

Verified Access endpoints

Transit gateways

Transit gateway attachments

Transit gateway policy tables

Transit gateway route tables

Transit gateway multicast

Metering policies

Traffic Mirroring

Mirror sessions

Mirror targets

Mirror filters

Network Manager

Cloud WAN

VPC IP Address Manager

You can visualize and monitor your Transit Gateway(s) from the AWS Network Manager. Register your Transit Gateway by creating a global network to get started.

Transit gateways (1/1)

Find transit gateway by attribute or tag

☒	Name	Transit gateway ID	Owner ID	State
☒	Org-TGW	tgw-0a82a4f5641ac89d3	986881030460	Available

Transit gateway: tgw-0a82a4f5641ac89d3 / Org-TGW

Details

Flow logs

Sharing

Tags

Details

Transit gateway ID

tgw-0a82a4f5641ac89d3

State

Available

Amazon ASN

64512

DNS support

Enable

Transit gateway ARN

arn:aws:ec2:us-east-1:986881030460:transit-gateway/tgw-0a82a4f5641ac89d3

Default association route table

Enable

Association route table ID

tgw-rtb-0cc039c89f90af102

Auto accept shared attachments

Disable

Owner ID

986881030460

Default propagation route table

Enable

Propagation route table ID

tgw-rtb-0cc039c89f90af102

VPN ECMP support

Enable

Description

Transit gateway CIDR blocks

Security Group Referencing support

Metering policy ID

CloudShell

Agent Toolkit for AWS

Feedback

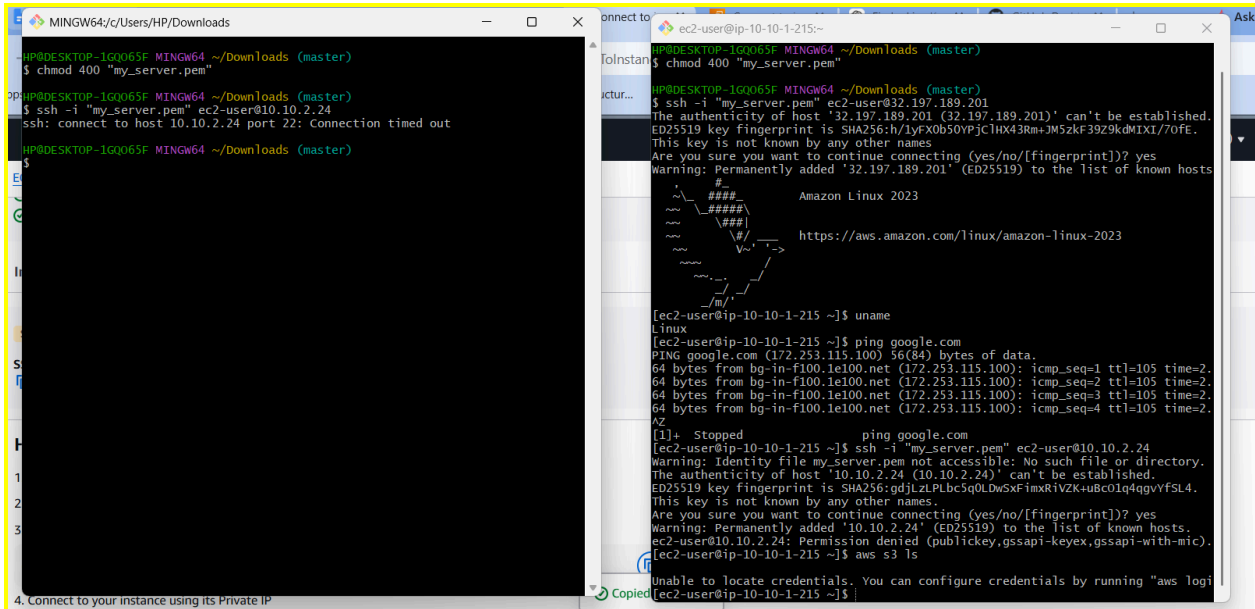
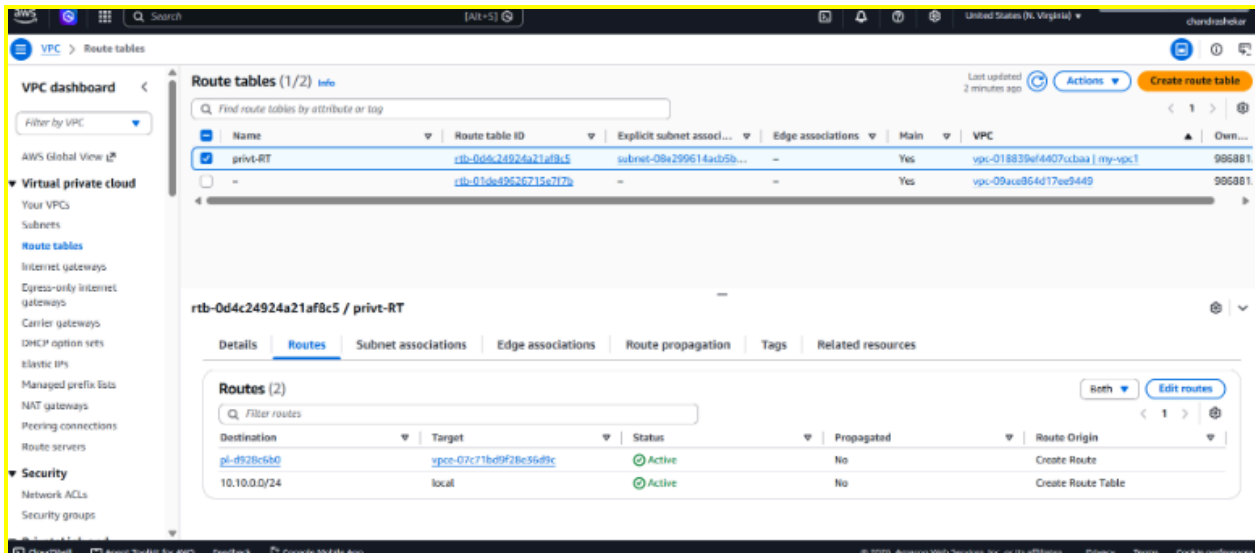
Console Mobile App

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4. Connect to your instance using its Private IP

```

-----END RSA PRIVATE KEY-----

[ec2-user@ip-10-10-1-215 ~]$ chmod 400 my_key
[ec2-user@ip-10-10-1-215 ~]$ scp -i my_key my_server.pem ec2-user@32.197.189.201:/home/ec2-user/
scp: stat local "my_server.pem": No such file or directory
[ec2-user@ip-10-10-1-215 ~]$ ls
my_key
[ec2-user@ip-10-10-1-215 ~]$ scp -i my_key my_server.pem ec2-user@32.197.189.201
cp: cannot stat 'my_server.pem': No such file or directory
[ec2-user@ip-10-10-1-215 ~]$ ssh -i my_server.pem ec2-user@10.10.2.24
Warning: Identity file my_server.pem not accessible: No such file or directory.
ec2-user@10.10.2.24: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-10-10-1-215 ~]$ cd ~/Downloads
ls *.pem
-bash: cd: /home/ec2-user/Downloads: No such file or directory
ls: cannot access '*.pem': No such file or directory
[ec2-user@ip-10-10-1-215 ~]$ cd ~/Downloads
-bash: cd: /home/ec2-user/Downloads: No such file or directory
[ec2-user@ip-10-10-1-215 ~]$ ssh -i my_key ec2-user@10.10.2.24

#_
~\##### Amazon Linux 2023
~\#####\
~\###|
~\#/_
~\V~' '-> https://aws.amazon.com/linux/amazon-linux-2023
~\m/'

[ec2-user@ip-10-10-2-24 ~]$ ping google.com
PING google.com (172.253.139.100) 56(84) bytes of data.

```

```

ec2-user@ip-10-20-1-66:~
HP@DESKTOP-1GQ065F MINGW64 ~/Downloads (master)
$ chmod 400 "my_server.pem"

HP@DESKTOP-1GQ065F MINGW64 ~/Downloads (master)
$ ssh -i "my_server.pem" ec2-user@18.234.112.249
The authenticity of host '18.234.112.249 (18.234.112.249)' can't be established.
ED25519 key fingerprint is SHA256:8m5pwxT5/LJZJydMqndpdjZNYg9yqghlopf7JkvzrrY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '18.234.112.249' (ED25519) to the list of known hosts.

#_
~\##### Amazon Linux 2023
~\#####\
~\###|
~\#/_
~\V~' '-> https://aws.amazon.com/linux/amazon-linux-2023
~\m/'

[ec2-user@ip-10-20-1-66 ~]$ vi my_key
[ec2-user@ip-10-20-1-66 ~]$ ssh -i my_key ec2-user@10.20.2.99
The authenticity of host '10.20.2.99 (10.20.2.99)' can't be established.
ED25519 key fingerprint is SHA256:q/feSgrGMwMqbbH5ujxq9z9KA13B77RKog+zREJysuE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.2.99' (ED25519) to the list of known hosts.
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@                WARNING: UNPROTECTED PRIVATE KEY FILE!                @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
Permissions 0644 for 'my_key' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key "my_key": bad permissions
ec2-user@10.20.2.99: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-10-20-1-66 ~]$

```

 ec2-user@ip-10-20-1-66:~

HP@DESKTOP-1GQ065F MINGW64 ~/Downloads (master)

```
$ chmod 400 "my_server.pem"
```

HP@DESKTOP-1GQ065F MINGW64 ~/Downloads (master)

```
$ ssh -i "my_server.pem" ec2-user@18.234.112.249
```

```
The authenticity of host '18.234.112.249 (18.234.112.249)' can't be established.
```

ED25519 key fingerprint is SHA256:8m5pwxT5/LJZJydMqndpdjZNYg9yqh0lop7JkvzrrY.

This key is not known by any other names

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

```
Warning: Permanently added '18.234.112.249' (ED25519) to the list of known hosts.
```

```
~\_#_
  ##_ Amazon Linux 2023
```

```

~ \#####\
~ \###|
~ \#/_____ https://aws.amazon.com/linux/amazon-linux-2023
~ V~' ->

```

```
[ec2-user@ip-10-20-1-66 ~]$ vi my_key
```

```
[ec2-user@ip-10-20-1-66 ~]$ ssh -i my_key ec2-user@10.20.2.99
```

```
The authenticity of host '10.20.2.99 (10.20.2.99)' can't be established.
```

ED25519 key fingerprint is SHA256:q/feSgrGMWmqbbH5ujxq9z9KA13B77RKog+zREJysuE.

This key is not known by any other names.

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

```
Warning: Permanently added '10.20.2.99' (ED25519) to the list of known hosts.
```

[illegible]

```
@          WARNING: UNPROTECTED PRIVATE KEY FILE!          @
```

[illegible]

```
Permissions 0644 for 'my_key' are too open.
It is required that your private key files are NOT accessible by others.
```

It is required that your private k

```
This private key will be ignored.
Load key "my-key": bad permissions
```

```
Load key "my_key": bad permissions
--3 user@10.20.3.00: Permission denied (publ
```

```
ec2-user@10.20.2.99: Permission denied (publickey,  
[ec2-user@ip-10-20-1-66 ~]$ ping 10.20.2.25
```

```
[ec2-user@ip-10-20-1-66 ~]$ ping 10.20.2.25
PING 10.20.2.25 (10.20.2.25): 56(84) bytes of data:
```

```
PING 10.20.2.25 (10.20.2.25) 56(84) bytes of data:
```

```
[1]: Stopped ping 10.30.3.25
```

```
[ec2-user@ip-10-20-1-66 ~]$ ping 10.30.3.35
```

```
lecz-user@p-10-20-1-66 ~]$ ping 10.30.2.23
PING 10.30.2.25 (10.30.2.25) 56(84) bytes of data:
```

```
PING 10.30.2.23 (10.30.2.23) 36(84) bytes of data:
^Z
```

```
[2]+  Stopped                  ping 10.30.2.25
```

```
[ec2-user@ip-10-20-1-66 ~]$
```